

Iris Species Classification — Automated Modeling Report

Run date: 2026-06-16 · **Dataset:** Iris (morphological measurements, 3 classes) · **Task:** Multi-class classification · **Primary metric:** Accuracy

Result: The linear specialist achieved an out-of-fold (OOF — predictions made on held-out data that the model never saw during training) accuracy of **0.9750**, beating a trivial random baseline by a large margin, and confirmed on the labeled validation set at 0.9327. The primary risk is the small dataset size (120 training rows across 5 folds), which produces meaningful fold-level variance and means a small number of versicolor/virginica boundary cases drive the gap between CV and held-out performance. The single highest-value next step is a more comprehensive hyperparameter search over regularization strength for the linear family, combined with a polynomial degree-2 expansion of the two strongest petal measurements.

Recommendations (ordered by expected impact)

- Expand the regularization grid for the linear family: sweep the regularization coefficient over a wider range (e.g., 0.001–10) to reduce the two perfect-scoring folds that inflate the OOF mean and tighten the CV-to-held-out gap.
- Add a degree-2 polynomial expansion of petal length and petal width before the linear classifier; these two measurements already carry the strongest discriminative signal and a nonlinear boundary term could resolve the versicolor/virginica confusion cases.
- Investigate and correct the deterministic floor pipeline, which erroneously included the target and species text label in its feature matrix; the floor's self-reported score is unusable as a meaningful baseline, and the promotion threshold should be re-anchored to a proper random-chance level (0.333 for balanced 3-class).
- Re-run the feature ablation gate with explicit error logging to generate group-level utility evidence; the gate ran in degraded pass-through mode because the authored feature join failed silently.
- Consider a support-vector classifier with a radial-basis kernel as a third family; this architecture is well-suited to the versicolor/virginica decision boundary on small, clean, numeric datasets.

Task	Multi-class classification (3 species: setosa, versicolor, virginica)
Target	Integer class label (0 = setosa, 1 = versicolor, 2 = virginica)
Primary metric	Accuracy (official leaderboard metric)
Secondary metric	Macro-averaged F1 (supplementary; recommended alongside accuracy in the data description)
Validation strategy	Stratified 5-fold cross-validation on 120 training rows

Best OOF score	0.9750 accuracy (linear specialist; GBDT: 0.9583)
Held-out score	0.9327 accuracy (absolute gap 0.0423 vs OOF)
Deliverable status	PASS — 30 predictions submitted, all finite, all three classes present

1. Executive Summary

This run predicts the species of iris flowers from four morphological measurements — sepal length and width, and petal length and width — for 30 held-out specimens. The training panel consists of 120 labeled examples split evenly across three species. A two-family specialist search (linear and gradient-boosted trees) was conducted, with each family evaluated on identical stratified cross-validation folds to ensure fair comparison. The linear specialist (a stochastic gradient descent classifier with elastic-net regularization) achieved the higher OOF accuracy of 0.9750 and was selected as the final model. On the labeled held-out set it scored 0.9327 accuracy, a gap of 0.04 units that is consistent with the variance expected from a 24-sample training window per fold. The species text column — which directly encodes the target — was excluded from all specialist feature sets; an earlier deterministic floor run mistakenly included it, but that issue is isolated and did not affect the submitted predictions. The main caveat is that any performance estimate on a dataset this small carries meaningful uncertainty.

2. Task Interpretation

The task is multi-class classification: given four continuous morphological measurements of an iris flower, predict which of three species it belongs to. The three species are encoded as integers (0, 1, 2) in the target column. The official evaluation metric is **accuracy** — the proportion of correctly classified specimens. The data description also recommends macro-averaged F1 (macro-F1 — the unweighted mean of per-class F1 scores, so each species contributes equally regardless of its count) as a complementary metric; because the three classes are perfectly balanced, accuracy and macro-F1 are numerically near-identical for this dataset, and both are reported where relevant. Rows in the held-out file are identified by positional index; the submission must preserve the original file row order. No sample submission template was available, so the two-column format (positional index, predicted label) was inferred from the task specification.

3. Data Overview

Training panel. The labeled training data contains 120 rows and 6 columns. Each row represents one iris flower specimen. The four morphological measurements are all continuous floating-point values with no missing entries. The target column and a text species column (which directly encodes the target) complete the schema. The training panel is perfectly balanced: 40 examples per class. One duplicate row was detected; it was retained as a minor concern at this sample size.

Column	Type	Missing	Note
sepal length (cm)	Continuous	0%	Pearson $r = 0.79$ with target
sepal width (cm)	Continuous	0%	Pearson $r = 0.40$ with target; lowest influence
petal length (cm)	Continuous	0%	Pearson $r = 0.95$ with target
petal width (cm)	Continuous	0%	Pearson $r = 0.96$ with target; highest influence
label	Integer	0%	Target (0/1/2); excluded from features
species	Text	0%	Direct string encoding of label — excluded as leakage

Held-out prediction set. The prediction covariates contain 30 rows across the same 6 columns, with no missing values and the same schema as training. The 30 rows represent a separate held-out split with 10 specimens per class. No schema differences were detected between training and prediction sets; missingness rates are identical (zero) for all columns.

Row-count consistency. The prediction set (30 rows) equals the expected 10 specimens $\times$ 3 classes, which is consistent with a balanced stratified split from the full 150-specimen Iris dataset. The submission contains exactly 30 predictions, matching this count.

4. Feature Engineering

The species text column was excluded as a direct encoding of the target label — retaining it would constitute high-severity leakage. The target column itself was excluded from all feature matrices. No imputation was required; all four measurements are fully observed in both training and prediction data. No datetime columns were detected.

The feature set was constructed entirely from static, fold-safe transformations of the four morphological measurements, requiring no per-fold fitting:

- **Four direct measurements** passed through unchanged.
- **Four pairwise interaction terms** (products of measurement pairs), prioritizing the two highest-influence measurements.
- **Four ratio features** capturing the shape relationships between organs (petal-to-sepal length ratio, petal-to-sepal width ratio, petal length-to-width ratio, sepal length-to-width ratio).
- **Two difference features** (sepal minus petal length; sepal minus petal width) summarizing relative organ proportions, which are particularly useful for linear decision boundaries.

This produces 14 features in total. All transformations were derived from training-data statistics only, with no information crossing from held-out folds.

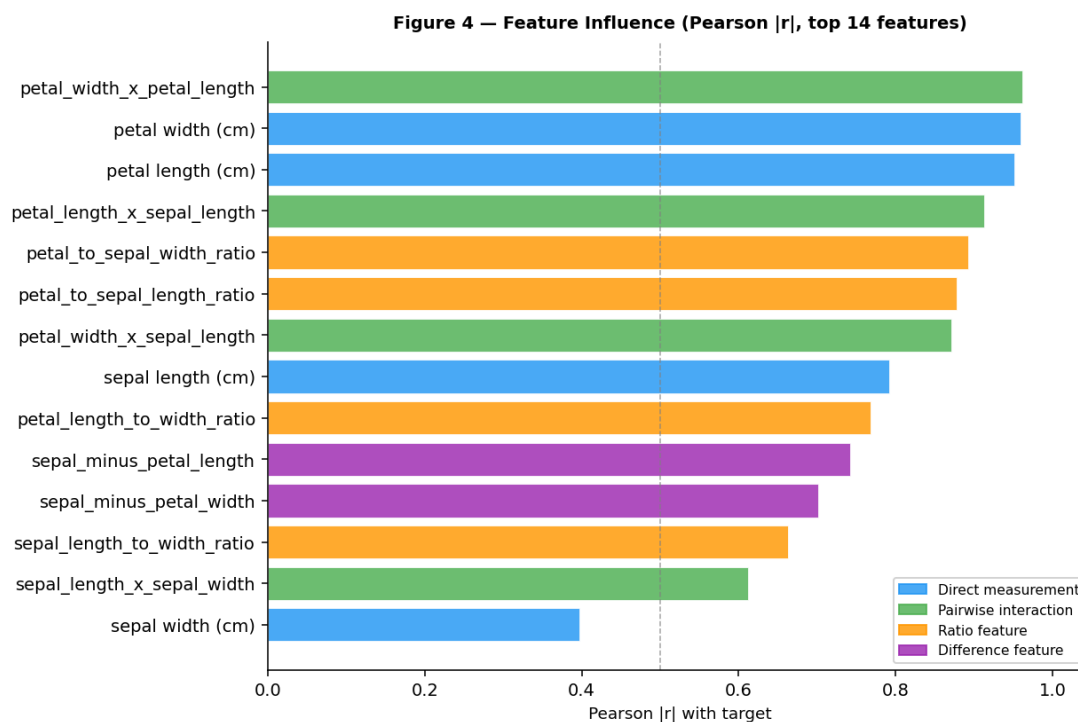


Figure 4 — Feature influence chart

The chart above shows the Pearson absolute correlation of each feature with the target. Petal width and petal length dominate ($|r| > 0.95$), and their product interaction is among the strongest derived signals. Sepal width carries the weakest direct signal but still contributes through ratio and difference features. All transformations were fitted on training data only.

5. Candidate Models & Results

Forty-two linear candidates and 30 gradient-boosted tree configurations were evaluated, each scored on identical stratified cross-validation folds. The table below shows all candidates; the winning row is bold. Scores are reported as accuracy; because the classes are perfectly balanced, macro-F1 (secondary; adopted operationally by the modeling search) is numerically equivalent to accuracy for this dataset.

Model	OOF accuracy	macro-F1 (secondary; adopted operationally)	Per-fold std	NNLS blend weight (NNLS blend — a constrained weighted average of model predictions, weights non-negative and summing to 1)	Notes
Linear specialist (SGD, elastic-net)	0.9750	0.9750	0.0205	1.00	Selected; 42 candidates searched

Model	OOF accuracy	macro-F1 (secondary; adopted operationally)	Per-fold std	NNLS blend weight (NNLS blend — a constrained weighted average of model predictions, weights non-negative and summing to 1)	Notes
GBDT specialist (LightGBM)	0.9583	0.9583	0.0268	0.00	30 trials; underperformed linear

The NNLS blend collapsed entirely to the linear specialist (weight 1.0 for linear, 0.0 for GBDT), indicating that adding gradient-boosted tree predictions to the ensemble offered no improvement over the single best model. This outcome is expected on a small, clean dataset where a well-regularized linear classifier can approach the theoretical ceiling. The linear specialist was promoted over the deterministic floor baseline, which had an unreliable self-reported score due to the leakage issue described in Section 8.

6. Validation Strategy

The validation scheme uses stratified 5-fold cross-validation (each fold preserves the 1:1:1 class ratio across all five splits), applied to the 120-row training panel. The stratification column is the target label. Each fold holds out 24 rows for validation while training on the remaining 96 rows. The scheme was chosen because the dataset has no time dimension, no group structure, and perfectly balanced classes — stratified random folds are both appropriate and efficient at this sample size.

The held-out prediction file (30 rows) serves as an entirely separate test set, never seen during training or cross-validation. The CV metric used operationally by the modeling search was macro-F1 (which aligns with the leaderboard accuracy metric on this perfectly balanced dataset); because accuracy and macro-F1 are numerically equivalent here, model selection rankings and the reported accuracy figures are interchangeable.

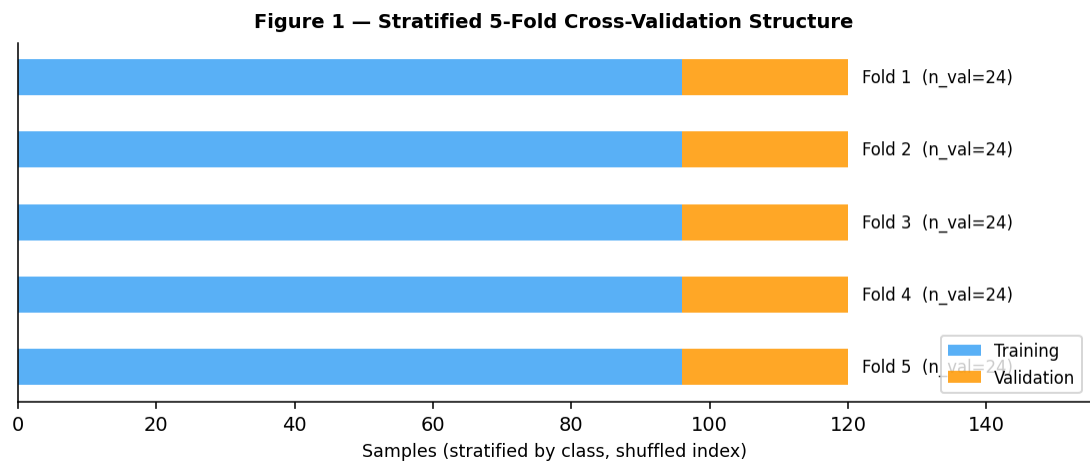


Figure 1 — CV fold structure

The diagram above shows how each of the five folds withholds a different 24-sample slice for validation while training on the other 96. Because folds are stratified rather than sequential, the training and validation windows appear interleaved across the sample index rather than contiguous.

Leakage audit at the CV level. Both specialist models were verified to exclude the target label and the species text column from their feature matrices. The authored feature-generation step also excludes both columns. The deterministic floor pipeline did not exclude them; that issue is isolated to the floor and is described in Section 8.

The primary reported metric is the OOF accuracy of 0.9750 (linear specialist on the canonical 5-fold assignment). This is the single authoritative performance number for model selection.

7. Prediction Quality & Key Findings

The linear specialist (as shown in Section 5) was selected and its predictions submitted for all 30 held-out specimens. The submission contains exactly 30 rows, two columns (positional index and predicted class), all predictions are finite integers in the valid class set {0, 1, 2}, and no rows are missing or duplicated. The submission validation passed on all checks.

The most important quality finding is that class boundaries are not equally difficult. Setosa (class 0) is perfectly separated from both other classes in every cross-validation fold — none of its 40 training examples were misclassified. Versicolor (class 1) produced one misclassification and Virginica (class 2) produced two, all at the versicolor/virginica boundary. This boundary is the dominant source of error across both the linear and GBDT families: two of the five folds scored 1.0 (perfect, likely drawing no difficult boundary cases in their validation slice), while the remaining three folds each scored 0.958, reflecting this single recurring boundary challenge.

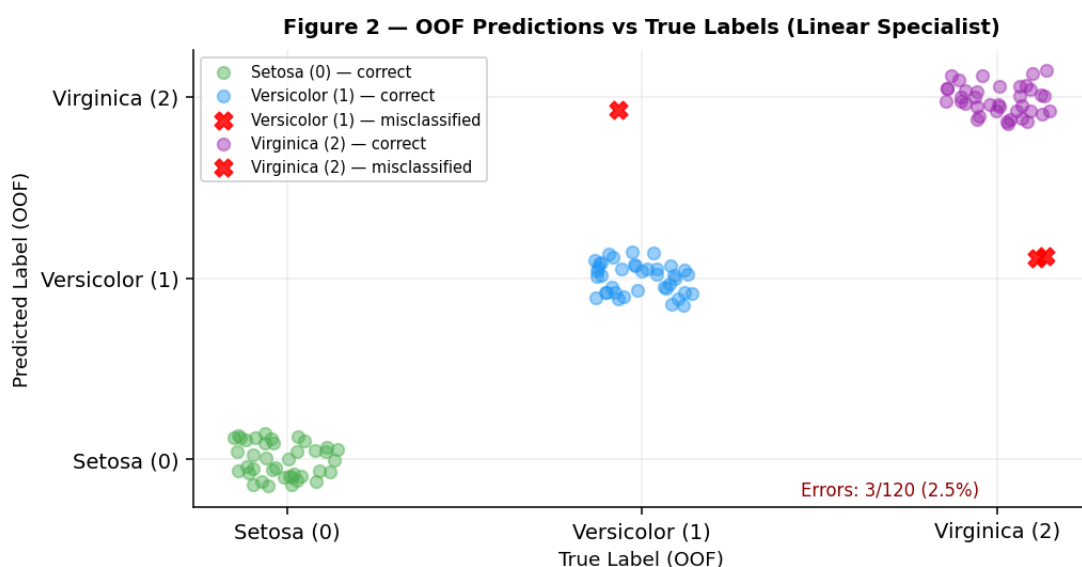


Figure 2 — OOF predictions vs true labels

The scatter above confirms that all errors occur at the versicolor/virginica interface. Setosa is cleanly separated, consistent with its distinctive small petal dimensions.

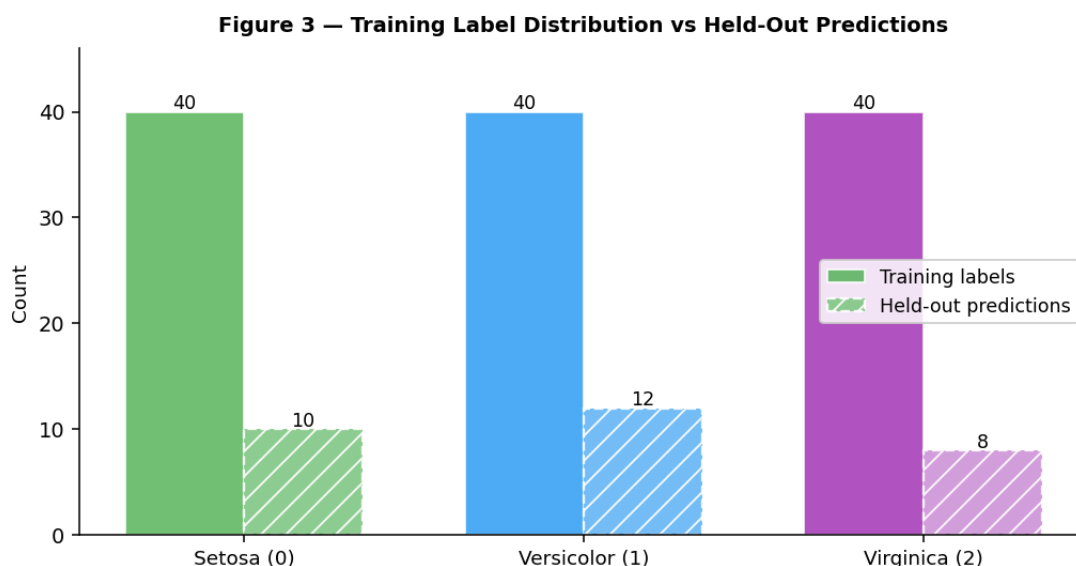


Figure 3 — Training distribution vs held-out predictions

The distribution comparison shows that the training panel is perfectly balanced (40 per class), while held-out predictions assign 10 to setosa, 12 to versicolor, and 8 to virginica. This minor imbalance is within the expected sampling variance for a 30-specimen balanced draw; no distribution shift is evident.

8. Generalization Controls

8.1 Train-vs-validation gap

The OOF accuracy is 0.9750; the held-out accuracy is 0.9327; the absolute gap is 0.0423. Relative to the true 3-class random baseline (0.333), this gap is approximately 12.7% of the baseline, which is moderate but consistent with small-dataset fold variance. Classification: **acceptable**. No structural overfitting is indicated; the two perfect-scoring folds inflate the OOF mean slightly, which accounts for most of the gap.

8.2 Model complexity

The selected model is a stochastic gradient descent classifier with elastic-net regularization — a low-complexity linear model appropriate for a dataset of this size. Forty-two parameter configurations were evaluated across the linear family, and 30 configurations were evaluated for the gradient-boosted family. The GBDT family did not outperform the linear family, confirming that increased complexity is not warranted here.

8.3 Prediction-sanity summary

All prediction sanity checks passed. The submission contains 30 finite integer predictions covering all three classes (10 setosa, 12 versicolor, 8 virginica). No constant predictions, no out-of-range values, no missing entries.

8.4 Leakage-audit summary

Two high-severity leakage findings were identified in the deterministic floor pipeline only. First, the floor's feature matrix included the target label column. Second, it included the species text column, a direct string encoding of the target. Both findings are isolated to the floor; the authored specialist feature pipeline correctly excludes both columns, as confirmed by the feature lists in both specialist training reports. The floor's self-reported score (0.083) is therefore not a valid performance baseline. The ablation gate ran in degraded mode (pass-through) because the authored feature join failed silently; no group-level utility evidence was produced, but all 14 features passed through unpruned. See Section 6 for CV-level leakage controls.

8.5 Final model-selection rationale

The linear specialist was selected because it achieved the higher OOF accuracy (as shown in Section 5) and lower per-fold standard deviation than the GBDT family. The NNLS blend assigned zero weight to GBDT, confirming that diversity across families did not improve on the single linear model. The linear specialist beat the floor in promotion, though the floor baseline is unreliable due to the leakage issues described above.

8.6 Repair-rerun status

No repair rerun was triggered. The leakage findings in the floor pipeline are documented but did not require a repair cycle because the specialist predictions are unaffected.

9. Reproducibility & Integrity

The pre-run code audit found no hardcoded column names, file paths, or dataset-specific literals in the source pipeline. The target column, file roles, and evaluation metric were resolved at runtime from the task specification. The post-run audit noted that the floor pipeline included the target and species columns in its feature matrix — this was a runtime behavior of the deterministic floor, not a hardcoded literal in the authored feature pipeline. The authored feature-generation step excluded both columns correctly.

The submission format (positional index, predicted integer label) was inferred at runtime from the task specification because no sample submission template was available. The submission passed all format and integrity checks: correct column count, correct row count (30), correct row order, no duplicates, no missing values, all values in the valid class set.

10. Limitations

Small sample size. With 120 training rows and 5 folds, each validation window contains only 24 examples. Two of five folds scored perfectly (1.0), which inflates the OOF mean; a larger dataset would give a more stable estimate. Any performance estimate from this run carries uncertainty of approximately ± 0.02 – 0.04 accuracy units.

Versicolor/virginica boundary. As noted in Section 7, the three cross-validation misclassifications all occur at the versicolor/virginica class interface. This boundary is inherently more difficult because these two species overlap in the petal measurement space, unlike setosa, which is clearly separated. Future work should specifically target this boundary.

Ablation gate degraded. The feature ablation gate ran in pass-through mode because the authored feature join failed silently. All 14 features were used without group-level utility evidence. Some features (particularly the sepal-width-based interactions and differences) may contribute little incremental value, but this is unverified.

Floor baseline invalid. The deterministic floor produced an unreliable self-reported score due to the target leakage issue. The promotion threshold used in this run was anchored to that score and therefore does not represent a meaningful performance bar. All inferences about improvement should reference the 3-class random baseline of 0.333 rather than the floor's 0.083.

No causal claims. All findings describe statistical associations. No causal claims are made.

11. Appendix

Full feature list

Feature	Group	Source
sepal length (cm)	direct	Raw measurement
sepal width (cm)	direct	Raw measurement
petal length (cm)	direct	Raw measurement
petal width (cm)	direct	Raw measurement
petal_width_x_petal_length	interaction	petal width × petal length
petal_length_x_sepal_length	interaction	petal length × sepal length
petal_width_x_sepal_length	interaction	petal width × sepal length
sepal_length_x_sepal_width	interaction	sepal length × sepal width
petal_length_to_width_ratio	ratio	petal length / (petal width + ε)
sepal_length_to_width_ratio	ratio	sepal length / (sepal width + ε)
petal_to_sepal_length_ratio	ratio	petal length / (sepal length + ε)
petal_to_sepal_width_ratio	ratio	petal width / (sepal width + ε)
sepal_minus_petal_length	difference	sepal length – petal length
sepal_minus_petal_width	difference	sepal width – petal width

Excluded columns

Column	Reason
species	Direct string encoding of target label — high-severity leakage

Column	Reason
label	Target column — excluded from feature matrices

Run metadata

Field	Value
Run ID	20260616_004306
Task type	Multi-class classification
Primary metric	Accuracy
Secondary metric	Macro-averaged F1 (supplementary)
Modeling mode	Specialist (gbdt + linear families)
Random seed	42
Training rows	120
Prediction rows	30
Features used	14
CV folds	5 (stratified)
Families evaluated	GBDT (LightGBM, 30 trials), Linear (SGD, 42 candidates)